TrashUQ

Leveraging Edge AI & Federated Learning
for Intelligent Waste Management

PRESENTATION
2024 | COMPANY/ UNIVERSITY

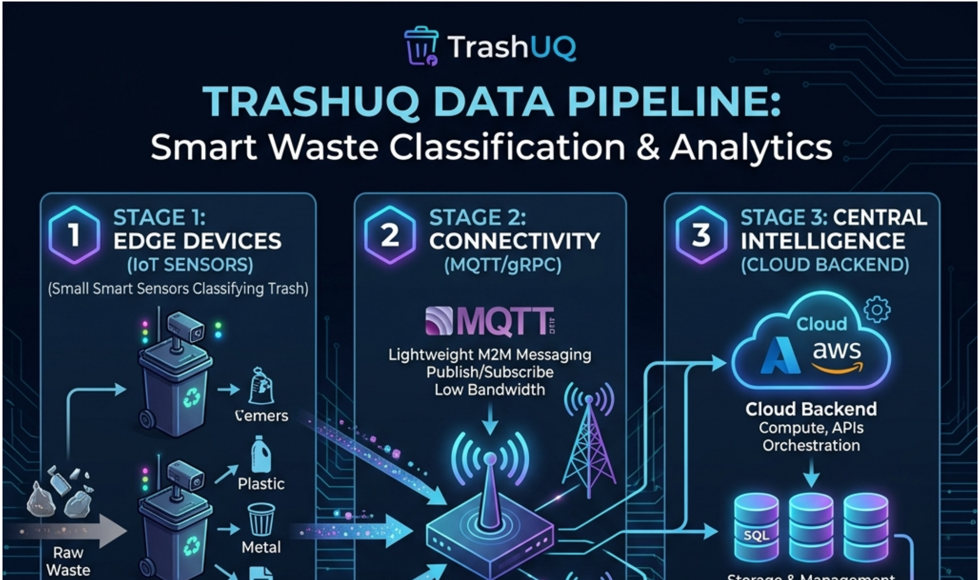
Executive Summary

- **TrashUQ** is an end-to-end platform for real-time trash classification monitoring.
- Integrates **Edge AI** on Arduino-class nodes with **Federated Learning** (FL).
- Combines MQTT telemetry, gRPC coordination, and a live Next.js dashboard.
- Successfully validated in a multi-node deployment with monotonic model versioning.

Key Features

- Real-time Observability
- Scalable FL Coordinator
- Concurrency Control
- TFLite Material Classification

System Architecture



Edge Intelligence

On-Device Inference

- Powered by **TensorFlow Lite (TFLite)**.
- Specialized for Arduino-class devices (UNO Q MPU).
- High-fidelity material classification.

Material Categories

- Cardboard
- Glass
- Paper
- Plastic

Real-time performance: <100ms inference on edge hardware.

Connectivity Telemetry

- **MQTT Protocol**: Lightweight pub/sub for real-time telemetry (status, metrics, events).
- **gRPC**: High-performance binary protocol for Federated Learning coordination.
- **Persistence**: PostgreSQL stores every message with full payload fidelity.

Data Integrity

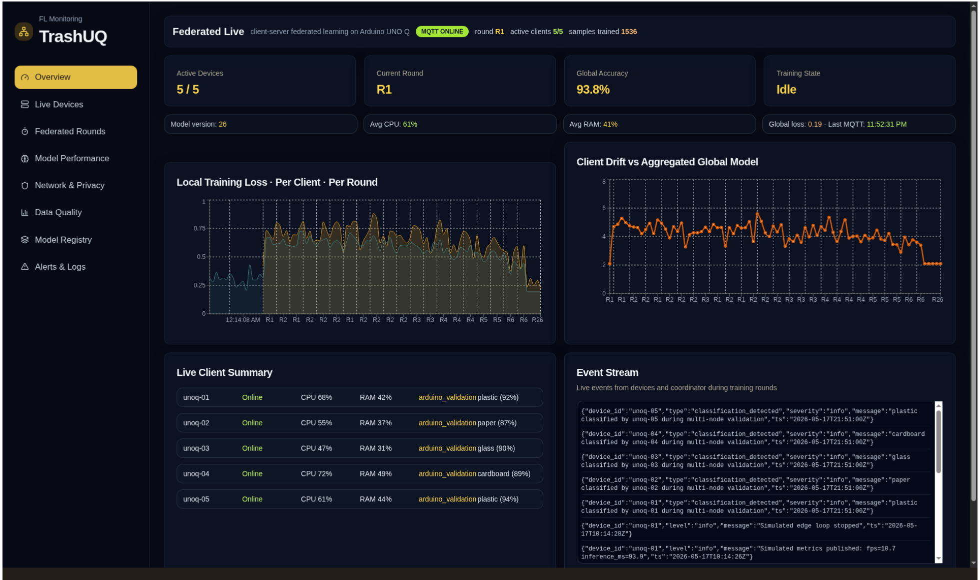
- Monotonic versioning
- Stale update rejection
- 0% dropped messages in validation

Federated Learning (FL) Lifecycle

Join Round → Get Global Model → Local Training → Submit Update

- **FedAvg Aggregation**: Weighted averaging based on sample count.
- **Coordination**: Handled via gRPC `SubmitUpdate` workflows.
- **Stability**: Proven convergence under Non-IID data partitions.

Live Monitoring Dashboard



Validation Highlights (Part A)

- **Deployment**: 2 Arduino nodes on `bepes-server`.
- **Activity**: 282 MQTT messages persisted.
- **Progress**: 26 FL rounds completed.
- **Success**: Model version advanced from 0 to 26.

Metric	**Value**
Aggregation Success	83.3%
Invalid Payload Rate	0%
Dashboard Lag	< 1s
Median Latency	$\sim$25ms

Scalability Performance (Part B)

- **Experiment**: Evaluated scaling from 2 to 20 clients.
- **Data**: Dirichlet Non-IID partitioning ($\alpha = 0.3$).
- **Accuracy**: Maintained **93-94% accuracy** across all scales.
- **Efficiency**: Communication cost scaled linearly with client count.

Result: Stable convergence even under highly heterogeneous data.

Conclusion & Future Roadmap

Achievements

- Proven edge-to-cloud FL pipeline.
- Robust concurrency and persistence.
- Deployed hardware validation.

Future Directions

- Production authentication IAM.
- Fleet management for 100+ nodes.
- Advanced FL analytics Differential Privacy.

The TrashUQ Team

Thank You!

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